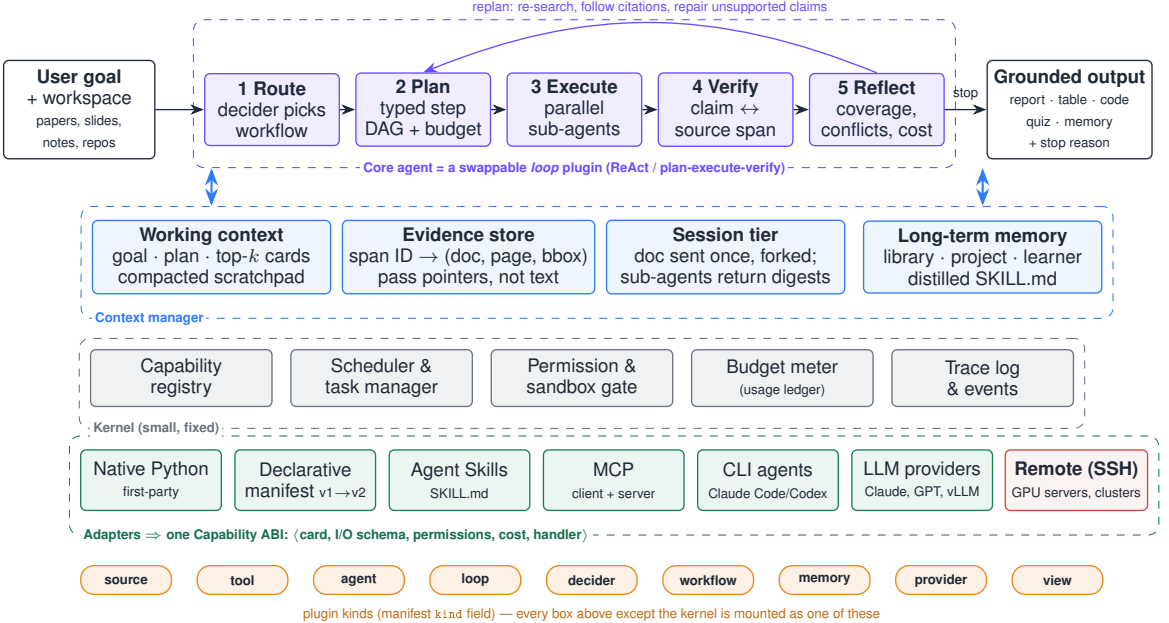

replan: re-search, follow citations, repair unsupported claims
User goal
+ workspace
papers, slides,
notes, repos
1 Route
decider picks
workflow
2 Plan
typed step
DAG + budget
3 Execute
parallel
sub-agents
4 Verify
claim ↔
source span
5 Reflect
coverage,
conflicts, cost
stop
Grounded output
report · table · code
quiz · memory
+ stop reason
Core agent = a swappable loop plugin (ReAct / plan-execute-verify)
Working context
goal · plan · top-k cards
compacted scratchpad
Evidence store
span ID → (doc, page, bbox)
pass pointers, not text
Session tier
doc sent once, forked;
sub-agents return digests
Long-term memory
library · project · learner
distilled SKILL.md
Context manager
Capability
registry
Scheduler &
task manager
Permission &
sandbox gate
Budget meter
(usage ledger)
Trace log
& events
Kernel (small, fixed)
Native Python
first-party
Declarative
manifest v1→v2
Agent Skills
SKILL.md
MCP
client + server
CLI agents
Claude Code/Codex
LLM providers
Claude, GPT, vLLM
Remote (SSH)
GPU servers, clusters
Adapters ⇒ one Capability ABI: ⟨card, I/O schema, permissions, cost, handler⟩
source
tool
agent
loop
decider
workflow
memory
provider
view
plugin kinds (manifest kind field) — every box above except the kernel is mounted as one of these